

Elucidating Superconductivity through the London and Ginzburg-Landau theories

Alexander Hu
8.06 term paper
(Dated: Spring 2023)

We explain the history, phenomenology, and modern applications of superconductors. We derive the London and Ginzburg-Landau theories for superconductivity and apply them to understand the Meissner effect, flux quantization, and difference between type-I and type-II superconductors.

I. INTRODUCTION

Superconductivity, a phenomenon characterized by the complete absence of electrical resistance and the expulsion of magnetic fields from the material, was discovered in 1911 by Dutch physicist Heike Kamerlingh Onnes. He noticed that below the liquefying temperature of helium, at around 4.2 K known as the critical temperature, the resistivity of mercury would suddenly drop to zero [1].

In 1933 German researchers Walther Meissner and Robert Ochsenfeld discovered the Meissner effect, where superconductors repel magnetic fields. In other words, when applying a magnetic field to a superconductor, the induced magnetic field cancels out the original magnetic field except on a thin surface layer, a behavior known as perfect diamagnetism [2].

To explain this effect, the London brothers proposed a phenomenological equation in 1935 and derive a key length scale known as the London penetration depth, which describes how fast the magnetic field decays inside a superconductor. Moreover, they showed that when there are holes in a superconductor, it can have a magnetic flux that is quantized in terms of the Dirac quantum. This was confirmed by Deaver and Fairbank in 1961 [3].

In 1950, Vitaly Ginzburg and Lev Landau developed a phenomenological theory to describe superconductivity, which is now known as the Ginzburg-Landau theory. The Ginzburg-Landau theory is a superset of the London theory. They used the idea of the order parameter ψ , first introduced by Gorter and Casimir, where $|\psi|^2$ is proportional to the density of the number of superconducting electrons. ψ can be thought of as a wavefunction, not for a single particle, but for the entire superconductor. It satisfies equations that resemble the Schrödinger equation and the equation for probability current for ordinary particles called the Ginzburg-Landau equations. These equations come from a form of free energy that Ginzburg and Landau derived from thermodynamics and electromagnetism. [4].

A key success of the Ginzburg-Landau theory is in describing superconductivity phase transitions. When increasing the magnetic field applied to a superconductor, there are two types of materials with different behaviors. In type-I superconductors, there is no intermediate state separating the transition from superconducting to normal state. To be clear, a normal state is where the material behaves like a regular conductor, where in particular

the total magnetic field inside the material approximately equals the applied magnetic field instead of zero like in superconductors. In type-II superconductors, there is an intermediate state called the mixed state, where the magnetic field penetrates in certain regions called vortices as soon as the magnetic field exceeds the lower critical field. When increasing the applied magnetic field in the mixed state, the vortices become more dense and numerous, until they start merging together. Once this happens, at the upper critical field, there is a transition to the normal state [1, 4].

Vortices are cylindrical tubes perpendicular to the surface of the superconductor extending into the superconductor. A vortex's radius is on the order of the coherence length, one of the two key length scales in the Ginzburg-Landau theory besides the London penetration depth. Another key property is that the flux through a vortex is quantized in terms of the Dirac quantum. So they can be thought of as a generalization of a hole in the London theory. The Ginzburg-Landau theory predicts that these vortices only form when the ratio of the coherence length and London penetration depth is greater than $\frac{1}{\sqrt{2}}$, which is the key property separating type-I and type-II superconductors. The theory can also be used to derive the two critical magnetic fields separating the fully superconducting, mixed, and normal states. Moreover, Alexei Abrikosov, nobel prize laureate, used Ginzburg-Landau theory to derive that vortices in materials occur in periodic lattices, often triangular, which was verified experimentally. These are called Abrikosov vortex lattices [3, 5].

In 1957, American physicists John Bardeen, Leon Cooper, and John Schrieffer developed the first widely-accepted theoretical understanding of superconductivity called BCS theory, winning them a Nobel prize in 1972. It explained superconductivity at a microscopic level and close to absolute zero temperatures [2, 6]. The theory is based on the notion that electrons in a superconductor can form bound pairs called Cooper pairs due to an attractive interaction mediated by phonons, quantized lattice vibrations in a solid [1]. BCS theory successfully explains various experimental observations, such as the isotope effect, where the critical temperature of a superconductor depends, inversely, on the mass of the isotopes when considering the same atom. Also, the BCS theory provides a quantitative description of the energy gap, a characteristic energy scale that separates the superconducting and normal conducting states [1]. Gorkov later

showed that the Ginzburg-Landau theory can be derived from the BCS theory in a limiting case, adding to the validity of the Ginzburg-Landau theory [7].

We will discuss London Theory in section II, using it to derive the Meissner effect. Then, in section III, we will derive the Ginzburg-Landau theory from the thermodynamics equations and explain the thermodynamics equations on an intuitive level. Using Ginzburg-Landau theory, we prove flux quantization in a hole of a superconductor and show intuitively why the size of the ratio of London penetration depth to coherence length relates to type-I versus type-II superconductivity. In section IV, we discuss the modern applications and areas of research in superconductivity.

II. LONDON THEORY

The London Theory already explains some phenomenology like the Meissner effect and flux quantization, albeit relatively non-rigorously. Also, the London penetration depth is one of the two important length constants in Ginzburg-Landau Theory, along with the coherence length. It is not completely rigorous. We present a direct way to derive it and the Meissner effect. Specifically, the Meissner effect is that the magnetic field decays exponentially inside a superconducting material, so that it is only non-negligible near the surface for a length-scale of order λ_L , the London penetration depth. Since λ_L is typically much smaller than the size of the superconducting material, it is common to say the magnetic field vanishes [4].

Recall that the order parameter of a superconducting material is the macroscopic wave function that is zero when the material is not superconducting and represents the number density of superconducting electrons $n_s = |\psi|^2$ otherwise [4]. We assume the magnetic field is zero and that the superconducting electrons are accelerated by a constant electric field. The superconducting current density is $j_s = -en_s v_s$. Hence we get

$$\frac{d\mathbf{j}_s}{dt} = \frac{n_s e^2}{m} \mathbf{E}. \quad (1)$$

By Maxwell's equations, using cgs units, we have $\nabla \times \mathbf{E} = -\frac{1}{c} \frac{d\mathbf{B}}{dt}$ and $\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{j}_s$. Taking curl of both sides and substituting in the Maxwell's equations, we define our London penetration depth

$$\lambda_L \equiv \frac{mc^2}{4\pi n_s e^2} \quad (2)$$

and get

$$-\lambda_L^2 \frac{d}{dt} \nabla \times (\nabla \times \mathbf{B}) = \frac{d\mathbf{B}}{dt}. \quad (3)$$

Since $\nabla \cdot \mathbf{B} = 0$, by curl identities the left hand side is $-\nabla^2 \mathbf{B}$, so

$$\lambda_L^2 \frac{d}{dt} \nabla^2 \mathbf{B} = \frac{d\mathbf{B}}{dt}. \quad (4)$$

Here the London brothers made the assumption that they could remove the time derivatives without leaving a constant behind [7]. This is motivated because if there was a constant solution for $\mathbf{B}$ this would violate the Meissner effect. Hence we get the key result

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda_L^2} \mathbf{B} \quad (5)$$

with exponentially decaying $\mathbf{B}$ on the order of λ_L . The solutions where $\mathbf{B}$ is exponentially growing are considered unphysical by the Meissner effect so they are discarded [7].

III. GINZBURG-LANDAU THEORY

A. Ginzburg-Landau Equations

The fundamental idea in Ginzburg-Landau theory is to write the free energy of the superconducting state relative to the normal state as a functional of the order parameter ψ . In thermodynamics, the free energy represents a system's ability to do work. The temperature T of our system is assumed to be held constant. The Landau free energy density is [8]

$$f_L = a(T)|\psi|^2 + \frac{1}{2}b(T)|\psi|^4 \quad (6)$$

where a and b are real coefficients that are functions of the temperature and assumed constant throughout our system. The motivation is that we are Taylor-series expanding the free energy as a function of the order parameter to two terms, keeping only even powers since odd ones can be not real, and having no constant term since we are considering the free energy relative to the normal state where $\psi = 0$ [4]. While $b > 0$ for thermodynamic stability, and for $T < T_c$, where T_c is the critical temperature, we have $a < 0$. At equilibrium, the Landau free energy will be minimized. This happens at

$$|\psi_\infty|^2 = -\frac{a}{b}, \quad f_L(|\psi_\infty|) = -\frac{a^2}{2b}. \quad (7)$$

From thermodynamics, we get at equilibrium

$$f_L = -\frac{H_c^2(T)}{8\pi}, \quad (8)$$

where $H_c(T)$ is the critical magnetic field, the maximum field one can apply before the superconductor becomes non-superconducting. We determine $H_c(T)$ by setting equal the thermodynamic results for the free energy of the system if it were superconducting and if it were not [5]. We thus have

$$\frac{a^2(T)}{b(T)} = \frac{H_c^2(T)}{4\pi}. \quad (9)$$

Now, Ginzburg's observation was that superconductors have charged particles, say electrons, so to include

these charged particles he added the gauge-covariant kinetic energy term appearing in the Schrodinger equation assuming the system order parameter ψ is like a particle of charge q and mass m [3, 9]. Later the values $q = -2e$ and $m = 2m_e$ where m_e is the mass of the electron were determined to work best experimentally and were also proved to be correct by the later BCS theory as they correspond to Cooper pairs [1]. Moreover we should take into account the magnetic field energy since it is non-zero. This explains the Ginzburg-Landau free energy [5]

$$f_{GL} = \int \left(\frac{1}{2m} |(-i\hbar\nabla - \frac{q}{c}\mathbf{A})\psi(\mathbf{r})|^2 + \frac{B^2}{8\pi} + a(T)|\psi(\mathbf{r})|^2 + \frac{1}{2}b(T)|\psi(\mathbf{r})|^4 \right) dV. \quad (10)$$

We minimize f_{GL} with respect to ψ^* to get

$$a\psi + b\psi|\psi|^2 + \frac{1}{2m}(-i\hbar\nabla - \frac{q}{c}\mathbf{A})^2\psi = 0. \quad (11)$$

This is analogous to the Schrodinger equation for a charged particle. We minimize f_{GL} with respect to $\mathbf{A}$ to get

$$-\frac{\hbar q}{mc} \left[\frac{1}{2i} (\psi^* \nabla \psi - \psi \nabla \psi^*) - \frac{q}{\hbar c} |\psi|^2 \mathbf{A} \right] + \frac{1}{4\pi} \nabla \times \mathbf{B} = 0 \quad (12)$$

Applying Maxwell's equation $\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{j}_s$ where $\mathbf{j}_s$ is superconducting current we get

$$\mathbf{j}_s = \frac{q\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) - \frac{q^2 |\psi|^2}{mc} \mathbf{A}. \quad (13)$$

Another way to view $\mathbf{j}_s$ is in terms of the gauge-covariant momentum $-i\hbar\nabla - \frac{q}{c}$:

$$\mathbf{j}_s = \frac{q}{m} \Re \left(\psi^* (-i\hbar\nabla - \frac{q}{c}) \mathbf{A} \psi \right). \quad (14)$$

The boundary conditions are that the component of the current perpendicular to the boundary vanishes, which holds if, for instance,

$$\hat{\mathbf{n}} \cdot (-i\hbar\nabla - \frac{q}{c})\psi = ir\psi \quad (15)$$

where r is some possibly-varying real constant for all points on the boundary [9].

A key sanity check for these equations is to consider the case when $|\psi|^2 = n_s$ is constant. Then, equation 13 will reduce to equation 5, the important equation in London theory, by taking curl of both sides.

B. Applications

1. Flux quantization

While a superconductor normally has negligible flux through it due to the Meissner effect as the magnetic

field decays exponentially starting from the value of the applied field at the surface. However, when the superconductor has holes, there could be magnetic field through the holes. It turns out that the flux through the hole is quantized, so long as the superconductor surrounding the hole is thick enough and the hole is large enough relative to the London penetration depth λ_L .

For clarity of explanation, we consider a cylindrical superconductor with inner radius R_1 , outer radius R_2 , and length ℓ , such that $R_1, R_2 - R_1, \ell \gg \lambda_L$. Suppose there is some magnetic field applied in the direction of the axis through the cylinder and the hole.

We first write the order parameter as $\psi = |\psi|e^{i\theta}$ where θ is the phase of ψ , which is a function of space and time, and substitute this into equation 12 to get

$$-\frac{\hbar q}{mc} \left(\nabla\theta - \frac{q}{\hbar c} \mathbf{A} \right) |\psi|^2 + \frac{1}{4\pi} \nabla \times \mathbf{B} = 0. \quad (16)$$

We draw a loop ∂D enclosing surface D centered at and perpendicular to the axis of the cylinder of radius $\frac{R_1+R_2}{2}$ and located at the middle of the cylinder. Since the distance from any point in D to the cylinder surface is much greater than λ_L , the magnetic field at all points infinitesimally close to ∂D is effectively zero by the Meissner effect. So for all points in ∂D , we have from equation 16

$$\nabla\theta = \frac{q}{\hbar c} \mathbf{A}. \quad (17)$$

Now we find the magnetic flux through D , Φ_D , is related to the phase change over one loop $\Delta\theta$:

$$\frac{\hbar c}{q} \Delta\theta = \oint_{\partial D} d\ell \cdot \mathbf{A} = \int_D d\mathbf{S} \cdot \mathbf{B} = \Phi_D. \quad (18)$$

The phase has to be consistent at the same point, so we have $\Delta\theta = 2\pi n$ for some integer n , whence

$$\Phi_D = \frac{2\pi\hbar c}{|q|} n \equiv \Phi_0 n, \quad (19)$$

where $\Phi_0 \equiv \frac{2\pi\hbar c}{|q|}$ is the Dirac flux quantum where we have $q = -2e$ in our formulation as explained earlier. Moreover, experiments by Deaver and Fairbank in 1961 measured Φ_0 enabling scientists to determine with more confidence that $q = -2e$ instead of $q = -e$, which people previously believed [3].

As the area of the hole $\pi R_1^2 \gg 2\pi R_1 \lambda_L$, the approximate area of the region of D where the magnetic field inside the superconductor is non-negligible, and the magnetic field only decreases inside the superconductor by Meissner effect, so Φ_D is essentially the flux through the hole.

2. Type-I vs Type-II superconductors

Type-I superconductors transition from superconducting to normal at the critical magnetic field $H_c(T)$, while

type-II superconductors have two critical magnetic fields $H_{c_1}(T), H_{c_2}(T)$, the lower and upper critical fields, between which lies the mixed state. In the mixed state, superconductors have regions known as domains or vortex structures where they are normal, and regions where they are still superconducting [4]. There is a smooth transition from the center of a vortex region, which is completely normal, to the surface of the vortex region that are superconducting. Moreover, the size of a vortex is on the order of the coherence length [4, 7]

$$\xi \equiv \sqrt{\frac{\hbar}{2m|a|}} \quad (20)$$

where a and m are those values defined in Ginzburg-Landau theory. This expression naturally appears when rescaling the Ginzburg-Landau equation 11 to be dimensionless [3]. The ratio

$$\tau \equiv \frac{\lambda_L}{\xi} \quad (21)$$

can be shown to represent whether the superconductor is type-I or type-II, depending on whether it is less than $\frac{1}{\sqrt{2}}$ or greater than $\frac{1}{\sqrt{2}}$ respectively [3].

To more rigorously understand this phenomenon, we analyze a simplified scenario. We consider the formation of a vortex in a superconducting material extending from $x = 0$ to $x = \infty$ with constant $a(T)$ and $b(T)$ throughout, adjacent to a normal material at $x = 0$. A magnetic field $\mathbf{H}_c$ in the x direction with same magnitude as the critical magnetic field of our system, as in equation 9, is being applied to our system. We further assume that the order parameter $\psi = \psi(x)$ varies only in the x direction, and that $\psi(0) = 0, \psi(\infty) = 1$, since the material is normal at $x = 0$ and superconducting infinitely far away from the normal material. We further simplify by setting $q = 0$, as vortex structures can still emerge without considering charge.

We want to see if the formation of the vortex is energetically favorable compared to the state of the type-I superconductor, where everywhere is superconducting. Our Ginzburg-Landau equation in equation 11 becomes

$$a\psi + b\psi|\psi|^2 - \frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = 0. \quad (22)$$

We are interested in temperatures less than the critical temperature T_c , where $a < 0$, and we make the substitution $\psi \rightarrow \sqrt{\frac{-a}{b}}\psi$ to get

$$-\xi^2 \frac{d^2\psi}{dx^2} - \psi + |\psi|^2\psi = 0. \quad (23)$$

Finally we assume $\psi(x) \equiv g(x)e^{i\theta}$ for a real constant phase θ and rearrange to get

$$\xi^2 \frac{d^2g}{dx^2} = \frac{\partial}{\partial g} \left(\frac{1}{4}(1 - g^2)^2 \right). \quad (24)$$

We solve this by multiplying both sides by $g'(x)$ and integrating, whence

$$\xi^2 \left(\frac{dg}{dx} \right)^2 = \frac{1}{2}(1 - g^2)^2 \quad (25)$$

as the constant is zero since $g'(\infty) = 0$. Integrating again and using boundary conditions we get

$$g(x) = \tanh \left(\frac{x}{\sqrt{2}\xi} \right). \quad (26)$$

This $g(x)$ is proportional to the original ψ so it is proportional to the number density of superconducting electrons. $g(x)$ rises steeply from 0 at $x = 0$ to nearly 1 at $x = 5\xi$. When $g = 1$, where $x = \infty$, this is the value of g when the material is superconducting, so for $x = 0$ to ξ , since $g < 1$, there is a depletion of superconducting electrons relative to the fully superconducting state. This is essentially a normal-like region, as in a completely normal region, the density of superconducting electrons is 0. So we have a vortex forming from $x = 0$ to $x = \xi$. Note we could say the vortex is from $x = 0$ to $x = 0.2\xi$, say, since the constant really does not matter as the size of the normal region with g less than some fixed constant is on the order of ξ , so we can set by convention the size of the vortex to be ξ .

Before we substitute our assumptions into equation 10, we must consider that the critical magnetic field $\mathbf{H}_c$ is a fixed external magnetic field being applied to our superconducting system. This means we must use the expression for the Ginzburg-Landau Gibbs free energy density g_{GL} which differs from equation 10 in that it has a $\frac{(B-H)^2}{8\pi}$ term instead of a $\frac{B^2}{8\pi}$ term, where H is the magnitude of the applied external magnetic field and B is the magnitude of the actual magnetic field inside the system.

$$g_{GL} = \int \left(\frac{1}{2m} |(-i\hbar\nabla - \frac{q}{c}\mathbf{A})\psi(\mathbf{r})|^2 + \frac{(B-H)^2}{8\pi} + a(T)|\psi(\mathbf{r})|^2 + \frac{1}{2}b(T)|\psi(\mathbf{r})|^4 \right) dV. \quad (27)$$

Hence the Ginzburg-Landau Gibbs free energy density per unit area σ_{GL} of our system is

$$\begin{aligned} \sigma_{GL} &= \int_0^\infty \left(\frac{\hbar^2}{2m} \left| \frac{d\psi}{dx} \right|^2 + a|\psi|^2 + \frac{1}{2}b|\psi|^4 + \frac{(B-H_c)^2}{8\pi} \right) dx \\ &= \frac{a^2}{b} \int_0^\infty \left(\xi^2 \left(\frac{dg}{dx} \right)^2 - g^2 + \frac{1}{2}g^4 \right) dx + \int_0^\infty \frac{(B-H_c)^2}{8\pi} dx. \end{aligned} \quad (28)$$

Suppose our superconductor was type-I. We know $\psi = \sqrt{\frac{-a}{b}}$ like in equation 7 because this minimizes the free energy, as the extra $\frac{\hbar^2}{2m} \left| \frac{d\psi}{dx} \right|^2$ term becomes 0 in the

prior equation when ψ is constant, so we get a Ginzburg-Landau free energy per unit area $\tilde{\sigma}_{GL}$ of

$$\tilde{\sigma}_{GL} = \int_0^\infty \left(-\frac{a^2}{2b} + \frac{H_c^2}{8\pi} \right) dx \quad (29)$$

The $\frac{H_c^2}{8\pi}$ term arises from the $\frac{(B-H)^2}{8\pi}$ term inside of equation 27 as we can approximate $B = 0$ inside of superconducting regions due to Meissner effect. Also, we can approximate $B = H_c$ inside of normal regions since this is a reasonable approximation for conductors [3].

The condition for formation of a vortex is if $\sigma_{GL} - \tilde{\sigma}_{GL} < 0$ as the equilibrium state is that of lower free energy, so we compute

$$\begin{aligned} \sigma_{GL} - \tilde{\sigma}_{GL} &= \frac{a^2}{b} \int_0^\infty \left(\xi^2 \left(\frac{dg}{dx} \right)^2 + \frac{1}{2}(1-g^2)^2 \right) dx \\ &\quad + \int_0^\infty \frac{(B-H_c)^2 - H_c^2}{8\pi} dx \\ &= \frac{a^2}{b} \int_0^\infty (1-g^2)^2 dx + \int_0^\infty \frac{(B-H_c)^2 - H_c^2}{8\pi} dx \quad (30) \\ &= \frac{a^2}{b} \int_0^1 \frac{(1-g^2)^2}{g'} dg + \int_0^\infty \frac{(B-H_c)^2 - H_c^2}{8\pi} dx \\ &= \frac{H_c^2}{8\pi} \frac{4\sqrt{2}}{3} \xi(T) + \int_0^\infty \frac{(B-H_c)^2 - H_c^2}{8\pi} dx, \end{aligned}$$

where we use equation 25 repeatedly and use equation 9 on the last line. Since B decays on the order of λ_L in superconducting regions, we can approximate the expression by letting $B = H_c$ for $x = 0$ to λ_L and $B = 0$ for $x = \lambda_L$ to ∞ [3] which gives

$$\sigma_{GL} - \tilde{\sigma}_{GL} = \frac{H_c^2}{8\pi} \left(\frac{4\sqrt{2}}{3} \xi(T) - \lambda_L \right). \quad (31)$$

For $\kappa \gg 1$, or $\xi \ll \lambda_L$, $\sigma_{GL} - \tilde{\sigma}_{GL} < 0$ so the system prefers to form a vortex, and vice versa for $\kappa \ll 1$. This argument shows that the tug-of-war between the loss of free energy due to magnetic field penetration on the order of λ_L and the increase in free energy due to the size of the vortex on the order of ξ is what determines whether vortices are energetically favorable to form versus remaining an ordinary superconducting state. By doing more exact calculations, $\sigma_{GL} - \tilde{\sigma}_{GL} = 0$ exactly when $\kappa = \frac{1}{\sqrt{2}}$. This argument can be extended to show that $\kappa = \frac{1}{\sqrt{2}}$ is the cutoff between type-I and type-II superconductors [5].

IV. CONCLUSION, APPLICATIONS, AND CURRENT PERSPECTIVES

Our two applications of the Ginzburg-Landau theory were the tip of the iceberg. Using these equations, we can derive many more results that explain experimental results well, ranging from equations for the upper and lower critical magnetic fields $H_{c2}(T) = \frac{\Phi_0}{2\pi\xi^2}$ and $H_{c1}(T) = \frac{\Phi_0}{4\pi\lambda_L^2} \ln(\kappa)$ where $\Phi_0 \equiv \frac{2\pi\hbar c}{|q|}$ is the Dirac quantum defined earlier and κ is the ratio of London penetration depth λ_L to coherence length ξ [7], to the formation of Abrikosov lattices [3]. These Abrikosov lattices are a very distinct pattern characterizing the structure of magnetic flux lines penetrating type-II superconductors in their mixed state and being able to be proven to exist by the Ginzburg-Landau theory is a great success [10].

Superconductors are important for modern technology. Superconducting magnets are used in the Large Hadron Collider at CERN to produce large currents without dissipating as much energy and to move and levitate high-speed Maglev trains [11]. In 1962, a Cambridge University graduate student Brian D. Josephson discovered Josephson junctions, where electrical current flows between two superconductors separated by an insulator, winning him the 1973 Nobel prize. This phenomenon is applied in devices like the SQUID, superconducting quantum interference device, the most sensitive instrument to detect magnetic fields known. Josephson junctions are also used to build quantum computers [2]. In magnetic resonance imaging (MRI) machines, a primary tool for medical diagnostics, approximately 35,000 out of the 50,000 machines worldwide use superconducting magnets, which offer higher resolution and faster scans. MRIs primarily use niobium titanium superconductors cooled with liquid helium [12].

A crucial area of research is in developing superconducting materials with higher, close to room temperature, critical temperatures. The US Department of Energy estimates that high-temperature superconducting wires to distribute energy in electrical grids could save the US hundreds of billions of dollars annually [11]. Since the discovery of superconductivity in 1911, more superconducting materials were found, such as niobium-nitride at 16 K in 1941, vanadium-silicon at 17.5 K in 1953, and the first commercial superconducting wire, an alloy of niobium and titanium in 1962. In 1987, the Fermilab Tevatron particle accelerator utilized magnets built out of copper-clad niobium-titanium [2]. In 1986, Alex Müller and Georg Bednorz, IBM researchers in Switzerland, won the Nobel prize the following year for creating a lanthanum, barium, copper, and oxygen ceramic compound that superconducted at a remarkably high temperature of 30 K. This spurred research and led to a 1987 breakthrough discovery by a University of Alabama-Huntsville team substituting yttrium for lanthanum to create Yttrium-Barium-Copper-Oxide (YBCO). This material achieves superconductivity at 92

K, warmer than the commonly available coolant liquid Nitrogen [2]. In 2015, scientists applied 155 gigapascals to Hydrogen sulfide breaking the 200 K barrier for the first time. Next, in 2020, University of Rochester physicist Ranga Dias and colleagues mixed hydrogen sulfide, hydride, and methane under 267 gigapascals creating a critical temperature of 288 K, achievable by a simple refrigerator [13]. However, this paper was retracted due to strong evidence of data fabrication and plagiarism [14].

Ginzburg-Landau theory has its shortcomings. High temperature superconducting CuO_2 was observed to have vortices arranged in pancakes of lattices stacked on top of each other in the direction of the applied magnetic field. These vortices were found to be able to be perturbed by thermal excitation unlike in Ginzburg-Landau theory. Moreover, in cuprate superconductors, the coherence length ξ can be on the order of one nanometer,

which is orders of magnitude smaller than the lengths in superconductors described well by Ginzburg-Landau theory. So it is found magnetic flux will concentrate at atomic size defects in CuO_2 like missing oxygen atoms or grain boundary defects. This occurrence is not part of the Ginzburg-Landau theory, which is expected because Ginzburg-Landau theory operates on a macroscopic level. There is room for new theories of superconductivity [10].

Acknowledgments

The author would like to thank Professor Maxim Metliski for teaching the class for which this paper was written and Mr. Ali Ghobadi and Mr. Rishab Parthasarathy for reviewing this paper.

-
- [1] Rafael M. Fernandes, *Lecture Notes: BCS theory of superconductivity*
 - [2] *The History of Superconductors* (superconductors.org)
 - [3] Daniel Arovas, *211B: Condensed Matter Physics Course Notes*
 - [4] James Charbonneau, *Introduction to the Ginzburg-Landau Equations* (Department of Physics and Astronomy, University of British Columbia, Vancouver, B.C. Canada, 2005)
 - [5] Phillip M. Duxbury, *Statistical Physics (PHY831): Part 4: Ginzburg-Landau theory, modeling of dynamics and scaling in complex systems* (2011)
 - [6] J. Bardeen, L. N. Cooper, J. R. Schrieffer, *Theory of Superconductivity* (Department of Physics, University of Illinois, Urbana, Illinois 1957)
 - [7] Masatsugu Sei Suzuki, *Ginzburg-Landau Theory for Superconductivity* (Department of Physics, SUNY at Binghamton, Binghamton, 2012)
 - [8] Todd Satogata, *The Ginzburg-Landau Theory*
 - [9] Barton Zwiebach, *Mastering Quantum Mechanics*
 - [10] R. P. Huebener, *The Abrikosov Vortex Lattice: Its Discovery and Impact*
 - [11] *Pulling together: Superconducting electromagnets* (CERN)
 - [12] Michael Parizh, Yuri Lvovsky, and Michael Sumption, *Conductors for commercial MRI magnets beyond NbTi: requirements and challenges*
 - [13] Ethan Siegel, *How Close Are We To The Holy Grail Of Room-Temperature Superconductors?*
 - [14] Tom Hartsfield, *A claimed superconductor "breakthrough" raises troubling questions*